

Exercise to laboratory 2:
The role of solid boundaries:
Evolution of an initial disturbance

*

April 24, 2026

Deadline: Monday May 18, 2026.

1 Theory (10P)

1.1 Coastal Kelvin Waves.

- i. Describe in your own words what coastal Kelvin waves are. (2P)
- ii. State the necessary assumptions to study coastal Kelvin waves. (1.5P)
- iii. Write down the solutions for the Shallow Water Equations for coastal Kelvin waves. (1P)
- iv. Describe the shape of the coastal Kelvin waves. (2P)

1.2 Equatorial Kelvin Waves

- i. Describe the assumptions to study equatorial Kelvin waves. (1.5P)
- ii. Write down the solution for the Shallow Water Equations for equatorial Kelvin waves. (1P)
- iii. In which direction do the equatorial Kelvin waves travel? (1P)

* Author: Verena Jung and Ezra Eisebrenner (2024), updated by Hanna Winge (2025)

2 Experiments

Set up the model as described in the laboratory description for Laboratory 2 and follow the instruction for the experiments.

2.1 Coastal Kelvin Waves (9P)

- i. Compute the phase speed from the position of the peak of the wave along the x-axis at $y=0$ and at different times. (1P)
- ii. Compare your results to the theory and discuss weaknesses of your results. (2P)
- iii. Provide plots showing the cross-section in the y-direction of your simulated coastal Kelvin waves. Ensure to include the analytical solution and the initial condition. (2P)
- iv. How is the result dependent on the resolution? (1P)
- v. Show plots of the cross section of these waves in x-direction for your experiments at different time steps.(1P)
- vi. Discuss changes of the wave with time and with resolution. (2P)

2.2 Equatorial Kelvin Waves (9P)

- i. Why do we want reduced gravity in this simulation? (1P)
- ii. Provide a plot of the horizontal cross-section at an appropriate time step and discuss the waves you can identify (include a cross-section through the equatorial Rossby waves in the same plot). (2P)
- iii. Provide a plot of the wave for multiple points in time (in x-direction, at $y=0$). (1P)
- iv. From that, compute the phase speeds and compare to theory. (2P)
- v. Provide a plot showing cross sections of the wave amplitudes in y-direction. Make sure to also include the initial condition and the analytical solution and compare your result to the theory. (2P)
- vi. Provide examples of where these waves be observed in the real ocean. (1P)

Good luck!